Cross-lingual Robust Concept Erasure via Token Embedding Editing

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Project Description:

Concept erasure aims to remove specific knowledge (e.g., entities such as "Harry Potter") from a language model while preserving general capabilities. Recent work shows that editing only MLP layers is often insufficient, as residual knowledge remains accessible through token embeddings. The EMBER framework addresses this by decomposing token embeddings into interpretable features using sparse matrix factorization and removing components associated with the target concept.

In this project, we focus on **cross-lingual robustness of concept erasure**. While EMBER operates on English token embeddings, it is unclear whether erased knowledge remains inaccessible when the same concept is queried in other languages.

We propose to study two main questions: (1) *Cross-lingual leakage*: does concept information re-emerge when prompts are translated into other languages, and how does this differ between embedding-based and MLP-based erasure methods? (2) *Multilingual extension*: can embedding-based erasure pipeline be extended to multilingual data, and do learned concept-related features remain shared across languages or become language-specific?

To address these questions, we will evaluate erasure performance across multiple languages using translated evaluation sets. We will compare recovery rates of erased concepts across languages and analyze how tokenization differences and language-specific representations affect erasure effectiveness. We further aim to extend the embedding editing pipeline to multilingual settings and evaluate whether this improves robustness.

Evaluation Plan:

We will measure erasure success using standard probing methods (e.g., model ability to answer concept-specific queries after erasure - open questions, multiple choice questions) across multiple languages. Metrics will include accuracy of concept recall (model's ability to regain concept recall after fine-tuning the original weights on a small amount of data), robustness under language translation, and comparison between embedding-based and MLP-based methods. We also intend to analyze qualitatively failure cases where concepts reappear.

Assumptions and Requirements:

The project assumes access to pretrained language models and the EMBER implementation, as well as GPU resources for running embedding factorization and evaluation. We will use publicly available translation APIs or open-source multilingual datasets to construct evaluation sets. No human annotation is required.

Expected Outcome:

We expect to characterize the limitations of current embedding-based erasure in multilingual settings and potentially propose simple extensions to improve robustness across languages. This will help understand whether and how concept representations are shared or language-dependent in embedding space.

Reference:

- C. Haya, O. Shafran, and M. Geva. *Don't forget your embeddings: Robust knowledge erasure via precise editing of embeddings.* arXiv preprint arXiv:2606.03695, 2026.